BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Paper 2: Programming Fundamentals

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What does insertion sort do with the next unsorted item?

2. Which is a benefit of a high-level language?
   A) Never needs translation
   B) Runs directly on every CPU
   C) Uses no programming rules
   D) Easier for humans to read

3. "A compiler must translate each unchanged source statement again every time the executable runs." Correct this. Explain why.

4. A programmer often changes and tests code. "An interpreter must make a standalone executable before running anything." Correct this. Why might an interpreter help here?

Today's objectives

Explain the core ideas of paper 2: programming fundamentals using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Sequence
Execute instructions in stated order.

Selection
Use Boolean conditions to take appropriate branch.

Iteration
Repeat a process using a controlled loop.

Modularity
Use subprograms and relevant data structures to avoid duplicated logic.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Sequence
Execute instructions in stated order.

Selection
Use Boolean conditions to take appropriate branch.

Iteration
Repeat a process using a controlled loop.

Modularity
Use subprograms and relevant data structures to avoid duplicated logic.

I Do - Worked model

Develop a program to input four numbers and output the smallest.

- Input first value into minimum.
- Loop three times inputting next values.
- If current value is smaller, replace minimum.
- Print minimum after all four values.

We Do - Guided practice

1. Plan a program to count how many of five marks are passes (>=50).

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the comparison that updates the smaller file size.

```
smallest = 9
newFile = 4
if ... then
smallest = newFile
endif
print(smallest)
```

You Do - Independent exam questions

Q2. Write a condition to test if x is even. [2 marks]

Q3. Explain why a loop is preferable to repeating the same five input lines. [2 marks]

Q4. State when a function is preferable to a procedure. [2 marks]

Exit Ticket - True or False?

Functions may reduce duplicated calculation logic.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Place the next item in the correct position in the sorted part. Shift larger items right.

2. D - Easier for humans to read

3. A compiler translates code before it runs. The executable can run again without translating unchanged source code.

4. An interpreter translates and runs code as it goes, giving quick testing feedback. A compiler creates code that can be run again and shared without translating it each time.

I Do - Explain the model

Input first value into minimum.

Loop three times inputting next values.

If current value is smaller, replace minimum.

Print minimum after all four values.

We Do - Model answers

1. Output passCount once at the end.

Code First - Answer

```
smallest = 9
newFile = 4
if newFile < smallest then
smallest = newFile
endif
print(smallest)
```

Expected output: 4

You Do - Self-mark

Q2: x MOD 2 == 0.

Q3: Reduces duplicated code and makes repetition count easy to change.

Q4: When a calculated value must be returned to the caller.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.